

Hybrid machine learning model for the prediction of anaemia

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ABSTRACT

In developing countries like Tanzania, despite the national intervention, the proportion of anaemic children aged 6–59 months is seen to be high, with a prevalence of 59 %. Traditional methods, such as examining paleness in the eyes and tongue, are commonly used, but are subjective and often lead to delayed or missed diagnoses. While existing Machine learning models have attempted to predict anaemia in children and offer improved accuracy, many rely on single-model strategies and a default threshold of 0.5, which tends to favour sensitivity over specificity, leading to a high number of false positives. The study used the supervised machine learning approach within the CRISP-DM framework. A stacked hybrid approach was used, integrating Random Forest (RF) and Artificial Neural Network (ANN) as base models, using XGBoost as a meta-learner. The model's performance was evaluated using metrics such as accuracy, sensitivity, specificity, precision, and Area Under the Curve (AUC) with 95 % confidence intervals (CIs) across thresholds of 0.35, 0.4, 0.45, and 0.5, optimized using Youden's J index. The hybrid model achieved balanced performance, especially at a 0.4 threshold with a sensitivity of 0.861 and a specificity of 0.880. Compared to standalone models, the hybrid approach outperformed in reducing false positives and false negatives, offering greater reliability and clinical safety. This study concludes that the stacking ensemble approach, along with threshold optimization, provides an effective solution for early detection of anaemia in children. Integration of hybrid Machine Learning models into Tanzania's health screening programs could improve child health outcomes.

1. Introduction

Anaemia refers to a condition characterized by a decrease in the number or quality of red blood cells, and remains a serious global health concern, particularly affecting young children, menstruating adolescent girls, and pregnant and postpartum women (Gardner et al., 2023). According to the World Health Organization (WHO), the proportion of anaemia varies across different groups worldwide, with 40 % of children aged 6–59 months, 37 % of pregnant women, and 30 % of women aged 15–49 years affected (Gardner et al., 2023). In developing countries like Tanzania, despite the national intervention, the situation is still critical, with 59 % of children in this group suffering from anaemia, according to the Tanzania Demographic and Health Survey (Ministry of Health (MoH)[Tanzania Mainland] et al., 2022).

Current clinical screening methods are often subjective and can lead to delayed or missed diagnoses, hindering timely intervention to eradicate the problem. To tackle this problem, there is an urgent need to improve the prediction methods that use modern data analytics and

machine learning techniques to enhance the prediction and timeliness of anaemia diagnosis in children aged 6–59 months.

Existing machine learning studies have limited performance metrics, such that the sensitivity and accuracy show higher performance than specificity, especially in an imbalanced dataset like the health dataset, which indicates many false positives, as many non-diseased (non-anaemic) children might be incorrectly classified as diseased (anaemic), leading to unnecessary treatment or misallocation of medical resources. Lower specificity indicates the possibility of missed cases. This is because most of these studies used the traditional default threshold of 0.5, which optimizes only one metric. Additionally, most of these studies relied on a single model strategy.

To address the mentioned challenges, the study proposes a novel Hybrid machine learning model using a stacking approach that combines Random Forest (RF), a tree-based model and Artificial Neural Networks (ANN), a deep learning model, (RF+ANN) with an XGBoost model, a meta-learner integrating with the threshold optimization using Youden's index to ensure a balanced sensitivity and specificity for the

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proper classification of the anaemic and non-anaemic children. The framework has not been previously reported in predicting pediatric anaemia, especially in Tanzania.

Additionally, this study benchmarked the hybrid model against several baseline and state-of-the-art approaches, such as LASSO, SVM, Naive Bayes, Random Forest, ANN, XGBoost, and Logistic Regression, using the same dataset, under the same experimental protocol. Results were reported with 95 % confidence intervals to ensure fair comparison. The statistical significance tests, such as McNemar's test for accuracy and DeLong's test for AUC, were performed to show the comparative performance between the proposed hybrid model and baselines.

2. Background review

Anaemia is a condition characterized by a lower-than-normal amount of red blood cells or a haemoglobin concentration that is <110 g/L at sea level, and hence leads to paleness, fatigue, dizziness, and difficulty breathing (Gardner et al., 2023). Bate and Rasmussen (2009) mentioned that Iron deficiency is the primary cause of anaemia in children due to insufficient iron intake, deficiencies in vitamins like vitamin A and vitamin B12, infections such as malaria, tuberculosis, HIV, and parasitic infections that impair nutrient absorption and metabolism, and inherited red blood cell disorders, including gene variants, and sickle cell that change the haemoglobin structure. If left untreated, anaemia directly affects the child's immune system, physical and cognitive development, and academic performance.

2.1. Factors associated with the cause of anaemia in children aged 6–59 months

Studies conducted by Kejo et al. (2018); Msaki et al. (2022); Said et al. (2021) reported that the factors such as the child's age, low birth weight, nonconsumption of iron-rich foods like meat, vegetables, and fruits, being a mother with no schooling, being a male child, being a child within the 6–59 months age, home births, maternal anaemia, child's stunting, and fever. These factors are categorized into Social, demographic, and health-related factors.

2.2. Feature selection techniques

Feature selection techniques are the methods applied to select or identify the subset of the most appropriate features from the dataset while leaving out the redundant or irrelevant ones to improve the model's interpretability, computational efficiency, and predictive performance (Laakel Hemdanou et al., 2024). There are three types of feature selection techniques: Filter methods, such as Chi-square tests, correlation, and mutual information; Wrapper methods, including Recursive Feature Elimination (RFE); and Embedded methods, including LASSO and Random Forest.

Most studies have used the filter method to select the best or significant features based on statistical relevance (Tesfaye et al., 2024; Uwizera et al., 2022; Zahirzada et al., 2023). Other previous studies have shown the superiority of the Embedded feature selection technique using random forest, as it selects the relevant features and is a capable solution to high-dimensional datasets that contain interactions among features of different types (Eldahshan et al., 2023; Saihood, 2022). In addition, using Random Forest as the feature selection technique, evaluates each feature's importance during the model training process, features with higher importance scores are selected for model building, and those with lower threshold scores are ignored (Pudjihartono et al., 2022).

2.3. Machine learning models in the prediction of anaemia

Recent studies have discovered the potential of Machine learning (ML) models that offer more objective and timely diagnoses compared to

traditional methods, since the proper intervention and diagnosis of anaemia in children 6–59 months can reduce the risk of anaemia, and enhance the detection of anaemia based on multiple risk factors (Uwizera et al., 2022). Several models, such as Random Forest (RF), XGBoost classifier, Support Vector Machine (SVM), Logistic Regression (LR), Naïve Bayes (NB), K-Nearest Neighbor (K-NN), Multi-Layer Perceptron (MLP), decision tree, and penalized regression models such as ridge, LASSO, and elastic net have been used to predict anaemia in children aged 6–59 months. A study by Uwizera et al. (2022) found that the XGBoost model was the best with an Accuracy of 85.21 %, Sensitivity of 98.71 %, Specificity of 71.70 %, F1-score of 86.97 %, and AUC of 98 %. Another study reported that RF outperformed the other models with an Accuracy of 86.4 %, AUC of 88.2 %, Precision of 85.0 %, and recall of 77.3 % (Zahirzada et al., 2023). (Tesfaye et al. (2024) showed that the LR model was the best model with an Accuracy of 66 %, Sensitivity of 82 %, Specificity of 42 %, and AUC of 69 %. Another study reported that LASSO had the highest performance with a Kappa of 28.56 %, an Accuracy of 64.29 %, a Sensitivity of 62.24 %, a Specificity of 66.32 %, a Positive predictive value of 64.71 %, a Negative predictive value of 63.89 %, and an F1 value of 63.45 %.

2.4. Evaluation metrics

A study conducted by Uwizera et al. (2022) considered accuracy, sensitivity, specificity, F1 score, and Receiver Operating Characteristics-area under the curve (ROC-AUC) as evaluation metrics. A study by Tesfaye et al. (2024) considered performance metrics such as Accuracy, Sensitivity, Specificity, and ROC-AUC. Accuracy, AUC, Precision, and Recall were the measures for assessing the model's performance (Zahirzada et al., 2023). Another study conducted to predict anaemia in children aged 6–59 months considered performance metrics such as Kappa, Accuracy, Sensitivity, Specificity, Positive predictive value, Negative predictive value, and F1 value for evaluation of the models (Meitei et al., 2022), while another study considered accuracy, sensitivity, and specificity as the evaluation metrics (Saihood, 2022).

This study used evaluation metrics such as accuracy, precision, sensitivity, specificity, and AUC. The 95 % confidence intervals (CIs) were computed using a non-parametric bootstrap procedure with 1000 resamples. Each resample was drawn with replacement from the test set, and the metric distribution across distribution resamples was used to calculate the percentile-based CI bounds. This approach avoids the assumption of normality and is widely used in classification performance evaluation (Skretteberg et al., 2024).

2.5. Threshold optimization

Threshold optimization in machine learning classification refers to the cut point or boundary used to assign a specific predicted class to each entity, and it is an important parameter when building and evaluating classification models, significantly affecting the model's performance (Guide, 2025). As is known, the traditional default threshold is 0.5 and is widely used in machine learning classification tasks, and the goal is to optimize a single performance metric, such as accuracy. However, some situations require more attention and are essential to maximize two or more performance metrics simultaneously to ensure balanced and robust classifier performance (Foody, 2023).

Lowering the threshold helps to balance the performance metrics. A study conducted by Foody (2023) reported that the optimal threshold of 0.3 provided a better balance among the three evaluation metrics compared to the default threshold of 0.5, the most significant metric, sensitivity, improved from 72.65 % to 84.65 %, and using the health dataset to identify the malignant tumors reported that the optimal threshold of 0.316 achieved the balance between sensitivity and specificity, indicating that the model identified a greater number of positive cases and reduced the number of false negatives. Another study conducted by Deo et al. (2024) reported that after applying the threshold

optimization, a notable improvement was observed in metrics such as precision and sensitivity. Therefore, the study applied the threshold optimization technique to optimize the sensitivity and specificity, as accuracy alone is insufficient to evaluate the model. This ensured that the model predicted well in detecting anaemic and non-anaemic cases.

2.6. Related works

Some more recent work uses advanced techniques such as spatial attention in images or aggregate model ranking (HBADR). A study conducted by [Asare et al. \(2023\)](#) considered features such as Hb values, age, sex, and remark (anaemic or non-anaemic) based on the Hb value captured with the corresponding images from the conjunctiva of the eyes, palpable palm, and fingernails and uploaded to the cloud storage (database) as the significant factors for anaemia detection in children using machine learning models such as Naïve Bayes, K-NN, SVM, Convolutional neural network (CNN), and decision tree. The CNN achieved a higher accuracy of 99.12 % compared to other models, indicating the power of CNN and the performance of the models, which explains that the non-invasive approach is an active instrument for anaemia detection.

Another study used the CP-AnaemiC dataset, the largest publicly available conjunctiva pallor-based dataset with annotated images of balanced anaemic and non-anaemic classes. Two representative tasks in anaemia diagnosis in children were performed, including Hb levels prediction and anaemia label classification. The study developed a deep neural network-based joint classification and regression learning framework, which simultaneously classifies anaemia and estimates Hb levels (g/dL) using the conjunctiva pallor images. The experimental results showed the effectiveness of the joint deep neural network in both the tasks of anaemia classification and Hb levels (g/dL) estimation ([Appiahene et al., 2023](#)).

Another study used the conjunctiva images to detect anaemia in children using machine learning models such as KNN, SVM, CNN, Naive Bayes, and Decision Tree. The CNN achieved a higher accuracy of 98.45 % indicating the importance of detecting anaemia in children using non-invasive techniques ([Asare et al., 2024](#)).

A study by [Saputra et al. \(2025\)](#) used an ensemble model that integrates Random Forest, XGBoost, and sea horse optimization techniques to determine the optimal number and depth of trees, hence improving accuracy and interpretability. The study also included an explainable artificial intelligence through SHAP analysis for transparency and trust in clinical decision-making. The study concluded that the Smart-Forest outperformed traditional models with an accuracy of 98.82 %, precision of 98.09 %, recall of 99.39 %, and F1-score of 98.72, indicating the reliability of the predictive power and the model's balanced nature.

Additionally, the study reviewed several related works to identify the Social, demographic, and health-related factors associated with the cause of anaemia in children aged 6–59 months, and the machine learning models used together with the evaluation metrics.

[Uwizera et al. \(2022\)](#) identified several key social, demographic, and health-related factors, including a child's age, malaria history, duration of breastfeeding, Residence, and Type of toilet facilities, as significant predictors of anaemia in children. The five machine learning models, such as RF, XGBoost, SVM, LR, and NB, were developed and evaluated. The XGBoost outperformed the other models with an Accuracy of 85.21 %, Sensitivity of 98.71 %, Specificity of 71.70 %, F1-score of 86.97 %, and AUC of 98 %. Thus, XGBoost was found to be reliable for detecting anaemic cases; however, it has a relatively low chance of identifying actual negative cases.

A study by [Tesfaye et al. \(2024\)](#) found that maternal and household factors such as maternal anaemia and weight, place of residence, and number of children under five as the most significant features for childhood anaemia. The models used included LR, RF, KNN, and DT. The LR model was the best model with an Accuracy of 66 %, Sensitivity of 82

%, Specificity of 42 %, and AUC of 69 %. This indicates that most of the anaemic children were correctly detected, while many non-anaemic children were incorrectly classified as anaemic. The model also had a fair chance (69 %) of correctly distinguishing between anaemic and non-anaemic children.

The study conducted in Afghanistan considered the child, maternal, and household-related factors such as maternal anaemia and weight, residence, and wealth status, as the most significant features for anaemia prediction in children aged 6–59 months. The ML models used were K-NN, MLP, NB, RF, and SVM. The RF outperformed the other models with an Accuracy of 86.4 %, an AUC of 88.2 %, a Precision of 85.0 %, and a recall of 77.3 %, indicating that the model was generally reliable in predicting anaemia status, correctly distinguished between an anaemic and a non-anaemic child, and the model correctly identified 77.3 % of the actual anaemic cases ([Zahirzada et al., 2023](#)).

In addition to model selection, several studies have emphasized the importance of hyperparameter tuning that yields the best performance for a machine learning model. [Trouvain et al. \(2020\)](#) urged that the random search gets more samples from the important parameters, while the grid search undersamples the space by repeating the sampling process for important parameters. A study conducted by [Ramzan et al. \(2024\)](#) reported using the GridSearchCV to enhance the performance of the models. Some studies that apply SVMs or Random Forests to classification tasks frequently report using grid search ([Chicco & Jurman, 2020](#)) and others, cross-validation-based tuning of hyperparameters ([Uwizera et al., 2022](#)). Similarly, the Artificial Neural Networks (ANNs) often require optimization of learning rate, number of hidden units, and dropout rates ([A Ilemobayo et al., 2024](#)). In this study, however, hyperparameter tuning was performed using a manual search and literature-guided approach, rather than automated search techniques such as grid search or randomized search, where values for each model were selected based on evidence from prior studies ([Goodfellow, 2015](#); [Jin et al., 2020](#)).

A study conducted by [Meitei et al. \(2022\)](#) considered the social, demographic, and health-related features such as the child's age and sex, number of living children, sanitation, and disposal of the youngest child's stool, as the most predictors of anaemia in children. The study used penalized regression models such as ridge, LASSO, and elastic net. LASSO achieved the highest performance with a Kappa of 28.56 %, an Accuracy of 64.29 %, a Sensitivity of 62.24 %, a Specificity of 66.32 %, a positive predictive value of 64.71 %, a negative predictive value of 63.89 %, and an F1 value of 63.45 %. Indicating that the model had a balanced but moderate ability to detect true anaemic and non-anaemic cases, and the Kappa value indicated fair agreement beyond chance, showing that while the model performs better than random guessing, there is still considerable room for improvement.

While these studies have mainly relied on the image-based CNN models ([Appiahene et al., 2023](#); [Asare et al., 2023, 2024](#)), used individual models such as Logistic Regression, Random Forest, and XGBoost ([Meitei et al., 2022](#); [Tesfaye et al., 2024](#); [Uwizera et al., 2022](#); [Zahirzada et al., 2023](#)). Additionally, [Saputra et al. \(2025\)](#) used advanced ensemble modelling through the Smart-Forest, combining RF and XGBoost with optimization and SHAP analysis for interpretability, but this design was restricted to the tree-based learners. On the other hand, this study advances the literature by combining the strengths of the two base learners, Random Forest (RF) and Artificial Neural Network (ANN), with XGBoost as a meta learner. This combination allows the complements of tree-based methods (robustness to feature interactions and non-linear relationships) and neural networks (capacity to capture complex patterns). This framework has not been reported previously in the area of predicting anaemia in children, specifically in Tanzania. Additionally, unlike the previous studies that mainly rely on the standard decision threshold of 0.5, this study incorporated the threshold optimization technique based on Youden's J index that optimizes the classification threshold to ensure a balanced trade-off between sensitivity and specificity, which is important in unbalanced datasets such as

pediatric anaemia and clinically relevant in healthcare settings.

2.7. Study gap

Despite these advancements in the related works, the gaps remain. Many existing machine learning models for the prediction of anaemia in children aged 6–59 months are faced with the problem of imbalance in performance metrics specifically in health dataset that might have the positive (anaemic) cases as the majority cases and negative (non-anaemic) cases as the minority cases, where the sensitivity shows the higher performance and specificity lower performance, representing many false positives as many non-anaemic children might be incorrectly classified as anaemic. This is because most of these studies used the traditional default threshold of 0.5, which favors only one metric on the cost of another metric (higher sensitivity and lower specificity). Additionally, most of these studies relied on a single machine learning model for anaemia prediction.

This study addressed these study gaps by integrating the strengths of more than one machine learning models, a Hybrid machine learning model using a stacking ensemble approach that combined Random Forest (RF), and Artificial Neural Network (ANN) (RF+ANN) with a meta-learner XGBoost model, as a result this approach increases the model's predictability power and generalizability with a threshold optimization approach that offered a balanced sensitivity and specificity that reduces the number of false positives (not anaemic children being classified as anaemic) and false negatives (anaemic children being classified as not anaemic).

3. Methods and tools

3.1. Study area

The study focused on urban and rural areas considering all usual residents in the households, including all women of reproductive age (15–49 years), men aged 15–49, and all children aged 0–59 months in the households across all regions of the Tanzanian Mainland and Zanzibar as covered by the Tanzania Demographic and Health Survey (2022). This survey was designed to provide representative estimates for the entire country and is conducted every five years.

3.2. Research plan

The Cross-Industry Standard Process for Data Mining (CRISP-DM) framework guides this study (Schröder et al., 2021). The framework consists of six phases: Business understanding, data understanding, data preparation, model building, and model evaluation. However, the study did not consider the sixth stage of deployment. The study was divided into three major parts (1) identifying the relevant social, demographic, and health-related factors associated with the cause of anaemia in children aged 6–59 months, (2) building a hybrid machine learning model that predicts anaemia among children aged 6–59 months, and (3) evaluate the performance of the proposed hybrid model. The hybrid model was evaluated across different performance metrics, including accuracy, precision, sensitivity, specificity, and AUC with 95 % confidence intervals (CIs) through different decision thresholds using Youden's J index. Additionally, statistical differences were assessed using McNemar's test to test whether one model outperforms the other, and DeLong's test was used to compare AUCs between ROC curves (Rainio et al., 2024).

3.3. Data sources

This study used secondary data from the Tanzania Demographic and Health Survey conducted in 2022 (Ministry of Health (MoH)[Tanzania Mainland] et al., 2022). This survey measures health issues in Tanzania.

3.4. Data preparation

Data was extracted from a dataset containing 10,783 records. After filtering the data to focus on the target age group (6–59 months), only 4366 records remained. Relevant features were recorded, and categorical values were mapped to facilitate model training. Missing values were checked and filled using K-Nearest Neighbours (KNN). Outlier values were checked using the Z-score method with a threshold of greater than positive or negative three Standard deviations. New features were created using the existing ones in the dataset, and those features used to create new features were dropped to improve the performance of the model.

3.5. Research tools

This study used the Python programming language (version 3.12.0), a popular choice in machine learning research due to its rich machine learning libraries. The use of Python, along with important libraries, includes Pandas for data cleaning, Matplotlib for data visualization, Seaborn for advanced statistical visualization, and Scikit-learn for building machine learning models (Sundaram et al., 2023).

4. Factors associated with the cause of anaemia in children aged 6–59 months

Since the dataset used contained both the numerical and categorical features; two different statistical tests were used: One way Analysis of Variance (ANOVA) was applied to numerical features such as the child's age in months (child_age), and mother's age (maternal_age) to compare mean differences between the anaemic and not anaemic groups; and the Chi-square test was applied for categorical features such as child's birth size (birth_size), and the child's stunting status (child_stunting) to examine whether the distribution of categories varied significantly across the outcome groups.

This study found that 14 features were significant in association with anaemia in children aged 6–59 months, while 13 features were not. These features are present in Table 1 when the value of "TRUE" in column Significant ($p \leq 0.05$) presents the statistically significant features, and the "FALSE" value presents not statistically significant features at 95 % Confidence Interval (CI) and $p\text{-value} \leq 0.05$.

Also, the features associated with anaemia in children aged 6–59 months that were directly linked to the child were the child's age in months (child_age), the child's diarrhea status in the last two weeks (haddiarr_last2w), the child's birth size (birth_size), the child's stunting status (child_stunting), the duration of the child's breastfeeding (breastfeed_duration), the child's fever Status in previous weeks (fever_last2w), child's sex (child_sex), and drugs for intestinal parasites status in previous 6 months (hadintestinal_drugs).

In addition, factors associated with anaemia in children, such as births in the last 3 years (birth_3yrs), the anaemia level of a mother (anemialev_mother), the mother's age (maternal_age), and whether a mother took iron supplements during pregnancy (irontab_preg), were associated with the mother's characteristics. The region (region) and source of drinking water (source_drinkwater) were related to anaemia in children aged 6–59 months, which was characterized by household factors.

5. Building a hybrid machine learning model

5.1. Feature selection

This study used random forest feature selection techniques to find the subset of features that improves the performance of the model. The random Forest model performed the feature selection technique through the repeated evaluation of random subsets of features at each tree split, with feature importance calculated by the average reduction in node

Table 1

Bivariate Analysis of anaemia status among children aged 6–59 months in all features.

Feature	Test	Test Value	p-value	Degrees of Freedom	Significant ($p \leq 0.05$)
child_age	ANOVA	276.0604452	4.97301E-60	1	TRUE
breastfeed_duration	Chi-Square	166.876991	5.79591E-37	2	TRUE
Region	Chi-Square	135.1576995	2.67066E-15	30	TRUE
birth_3yrs	ANOVA	60.10491563	1.12891E-14	1	TRUE
anemialev_mother	Chi-Square	66.67842743	2.19416E-14	3	TRUE
maternal_age	ANOVA	34.99417735	3.57927E-09	1	TRUE
birth_size	Chi-Square	41.05982977	2.61224E-08	4	TRUE
haddiarr_last2w	Chi-Square	24.76108869	6.48944E-07	1	TRUE
child_stunting	Chi-Square	19.81678987	8.52307E-06	1	TRUE
irontab_preg	Chi-Square	19.12579869	1.22379E-05	1	TRUE
child_sex	Chi-Square	11.36612719	0.000747957	1	TRUE
fever_last2w	Chi-Square	6.845189627	0.008888026	1	TRUE
source_drinkwater	Chi-Square	30.33103629	0.016359529	16	TRUE
hadintestinal_drugs	Chi-Square	4.975024286	0.025715843	1	TRUE
toilet_facility	Chi-Square	22.3364693	0.050362718	13	FALSE
mother_occupation	Chi-Square	19.37233858	0.054728693	11	FALSE
twin_child	Chi-Square	4.878357114	0.087232479	2	FALSE
mother_ednlevel	Chi-Square	4.765344594	0.189808533	3	FALSE
hhhead_sex	Chi-Square	1.437785812	0.230497981	1	FALSE
wealth_index	Chi-Square	4.425096063	0.351521772	4	FALSE
hhmember_no	ANOVA	0.486019401	0.485748229	1	FALSE
hadvitamin_a	Chi-Square	0.414922953	0.519481325	1	FALSE
less5_childhh	ANOVA	0.274637237	0.60026648	1	FALSE
marital_status	Chi-Square	2.796267371	0.731359942	5	FALSE
Wasting	Chi-Square	0.115905211	0.733518767	1	FALSE
residence	Chi-Square	0.03740724	0.846638265	1	FALSE
antenatal_visits	ANOVA	0.028749427	0.865367326	1	FALSE

impurity (Gini index) across all trees. Thus, ranking features according to their contribution to the model performance. The feature importance score, as indicated by the threshold line (red dotted line) from the chart, shows the threshold point of importance > 0.02 . Features to the left of the line have importance scores below the threshold, and are not considered, while those to the right of the threshold line score with higher bars have the more important features' scores and were considered for model building, as shown in Fig. 1. After important features have been selected through the RF feature selection technique, the two base models, RF and ANN, were then trained separately.

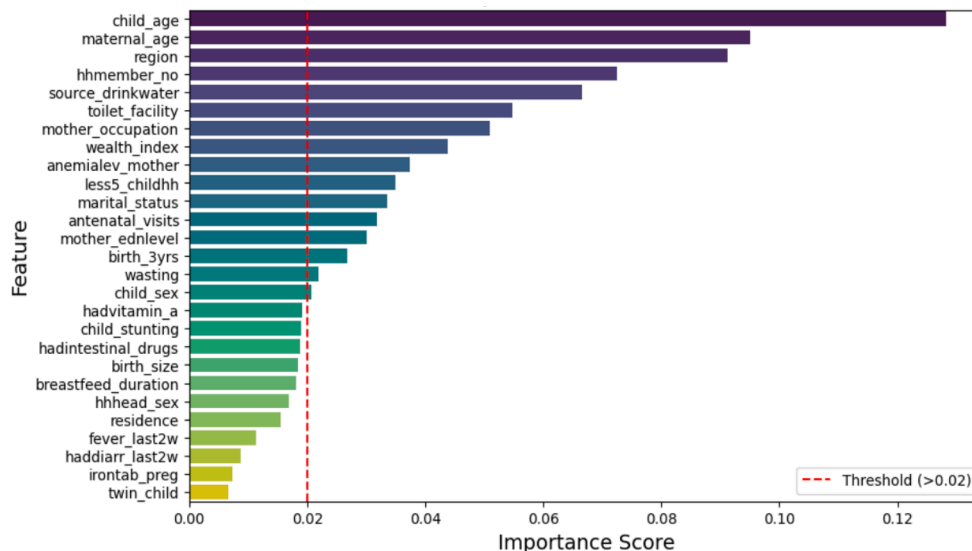
5.2. Hyperparameter tuning

The hyperparameters were carefully selected and tuned to ensure the robustness and generalizability of all models, RF, ANN, and Hybrid Stacking model (RF+ANN with XGBoost as the meta-learner). In this

study, hyperparameter tuning was performed using a manual and literature-guided approach. The hyperparameter values for each model were selected based on the evidence from the prior studies and empirical testing to ensure both model interpretability and generalization.

5.2.1. Tuning RF model

The ensemble model, Random Forest, was trained using 500 decision trees to enhance the generalization and reduce model overfitting. A study by Jin et al. (2020) reported that the generalization error decreases as the number of trees in the forest increases. The maximum tree depth was set to 10. To control overfitting, the minimum number of data points needed to split was five split samples or three leaf samples. Even more important, the class weights were set to be balanced, automatically adjusting the class weights for imbalanced class distributions.

**Fig. 1.** Feature Importance score with the Threshold Line.

5.2.2. Tuning the ANN model

The second Base model, Artificial Neural Network (ANN), was first trained before being evaluated using different evaluation metrics. The study used three hidden layers with 128, 64, and 32 neurons to train the ANN model. This is known as a pyramidal structure and is widely used in deep learning as it allows feature abstraction. A study conducted by Goodfellow (2015) highlighted that deeper networks with increased depths and decreasing layer sizes are active and successful for abstract feature extraction and enhance the generalization, as the first layer identifies general patterns and the other layers extract more complex information. The ReLU activation was used, which is beneficial for deep learning. Additionally, Adam optimizer for adaptive gradient method, and finally, early stopping was applied to stop training if no improvement was made to prevent model overfitting. Collectively, these parameters allowed the ANN to learn meaningful feature representations while maintaining generalizability capacity.

5.2.3. Tuning the hybrid model

This hybrid model was formed using two steps: first, getting the output probabilities from both RF and ANN, then cross-validation stacking was performed to avoid model overfitting. The data were then split into five folds or subgroups of about equal size using a stratified K-fold cross-validation, confirming that each fold contains the same proportion of the observations.

The second stage involved a meta learner, XGBoost model, which was finally trained to learn how best to combine the RF and ANN predictions for the Hybrid model with carefully tuned parameters. The meta-model was tuned using 500 trees, with 80 % of features per tree to maximize generalization, a small learning rate of 0.01 for more precise boosting, together with a maximum depth of 4, and 80 % of the data per tree to prevent overfitting.

5.3. Model development

From the dataset, 70 % of the data was used to train the models, and 30 % was used to evaluate them. The stratified splitting was applied to preserve the ratio of anaemic and not anaemic cases in both sets. The base (RF and ANN) models were first trained separately using the original training set (X_{train} and y_{train}), the hybrid (RF+ANN) model was then formed using two steps: first, getting the output probabilities from both RF and ANN, and second, training a meta-learner (XGBoost model) on these probabilities as input features to make the final prediction for the hybrid model. The mathematical formulation of the model is presented in the following steps.

Step 1: Base Models

Let

$X \in R^{n \times p}$ Is the matrix of a real dataset of the size $n \times p$, where n is the number of children (row), and p is the number of features (columns), and $y \in \{0, 1\}^n$ Is the binary output vector of length n (number of children or row) with values of 0 and 1 for a child, where 0 = not anaemic and 1 = anaemic.

The 2 base models were trained, as shown the following equations.

$$f_{RF} = X \rightarrow [0, 1] \quad (1)$$

The random forest model (f_{RF}) was trained on the input features to predict the chance of whether a child is not-anaemic (0) or anaemic (1).

$$f_{ANN} = X \rightarrow [0, 1] \quad (2)$$

The Artificial Neural Network (f_{ANN}) model was also trained on the input features to predict the probability of whether a child is not-anaemic (0) or anaemic (1).

$$P_{RF} = f_{RF}(X) \quad (3)$$

The output from (i) is treated as the probability from the random forest (RF) model, which corresponds to the chance of a child being anaemic.

$$P_{ANN} = f_{ANN}(X) \quad (4)$$

The output from the second base model in (ii) is treated as the probability from the Artificial Neural Network (ANN) model.

Step 2: Meta-Feature Construction

$$P_{meta} = [P_{RF}, P_{ANN}] \quad (5)$$

P_{meta} The meta-feature vector, which was obtained by combining the outputs from the base models into a 2-dimensional feature vector, the RF, and the ANN probabilities

Step 3: Meta-Model XGBoost

$$\hat{p} = f_{XGB}(P_{meta}) \in [0, 1] \quad (6)$$

The third step was training a meta-learner model, **XGBoost**, f_{XGB} using the meta feature vector, P_{meta} as an input, to get the final predicted probability from XGBoost, $\hat{p}$, which again contains the values of 0 = not anaemic and 1 = anaemic.

Step 4: Threshold Optimization using Youden's Index

$$J(\tau) = \text{Sensitivity}(\tau) + \text{Specificity}(\tau) - 1 \quad (7)$$

Lastly, instead of using the default threshold $\tau = 0.5$, the threshold optimization process using Youden's index was then done to get the optimal threshold (τ^*) to balance sensitivity and Specificity to ensure the balance between Sensitivity and Specificity.

Then the optimal threshold (τ^*) is given as:

$$\tau^* = \arg \max_{\tau \in [0, 1]} J(\tau)$$

Where the optimal threshold (τ^*) is the value of τ that maximizes $J(\tau)$, ensuring the best trade-off between sensitivity and specificity.

The final classification rule is given as:

$$\hat{y} = \begin{cases} 1 & \text{if } \hat{p} \geq \tau^* \\ 0 & \text{elsewhere} \end{cases}$$

Where:

$\hat{p}$ Is the final anaemia prediction, which is either 0 = not anaemic or 1 = anaemic. The child is predicted as anaemic (1) if the final prediction, $\hat{p}$ is greater than or equal to the optimized threshold, τ^* . Otherwise, the child is predicted as not-anaemic (0).

In addition, the proposed Aneamia Prediction Pipeline using a stacked ensemble hybrid (RF+ANN) with XGBoost as a meta learner machine learning model is presented in Fig. 2.

6. Results and discussions

6.1. Results

Results of this study are presented in the following subsections.

6.1.1. Performance of individual models across the threshold

Table 2 shows the performance of the evaluation metrics across different thresholds (0.35, 0.4, 0.45, and 0.5) for all three models, the

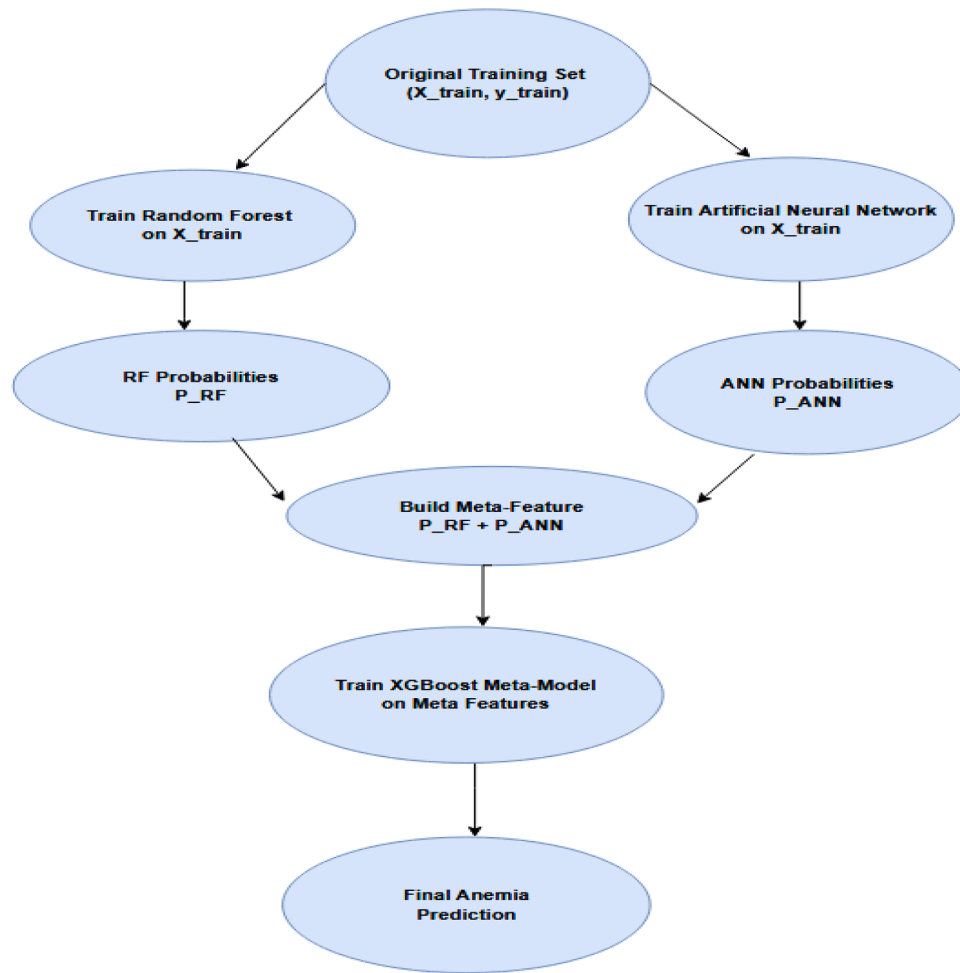


Fig. 2. The proposed Anaemia Prediction Pipeline using a stacked ensemble hybrid (RF+ANN) machine learning model.

Table 2

Performance Metrics of the three Models across Thresholds.

Model	Threshold	Accuracy (95 % CI)	Precision (95 % CI)	Sensitivity (95 % CI)	Specificity (95 % CI)	AUC (95 % CI)	Youden (95 % CI)
ANN	0.35	0.855 (0.834–0.874)	0.782 (0.747–0.820)	0.836 (0.801–0.867)	0.866 (0.840–0.890)	0.934 (0.920–0.946)	0.702 (0.656–0.742)
	0.4	0.860 (0.840–0.880)	0.800 (0.764–0.837)	0.823 (0.786–0.855)	0.881 (0.858–0.904)	0.934 (0.920–0.946)	0.704 (0.660–0.746)
	0.45	0.861 (0.839–0.880)	0.812 (0.775–0.848)	0.805 (0.766–0.839)	0.893 (0.869–0.915)	0.934 (0.920–0.946)	0.698 (0.650–0.738)
	0.5	0.861 (0.839–0.880)	0.822 (0.787–0.858)	0.789 (0.750–0.822)	0.902 (0.879–0.924)	0.934 (0.920–0.946)	0.691 (0.644–0.731)
Hybrid	0.35	0.871 (0.852–0.890)	0.794 (0.759–0.831)	0.874 (0.841–0.902)	0.870 (0.845–0.892)	0.953 (0.941–0.963)	0.744 (0.702–0.782)
	0.4	0.873 (0.853–0.892)	0.805 (0.771–0.840)	0.861 (0.827–0.890)	0.880 (0.856–0.902)	0.953 (0.941–0.963)	0.741 (0.697–0.778)
	0.45	0.875 (0.856–0.893)	0.822 (0.789–0.858)	0.841 (0.805–0.872)	0.895 (0.874–0.917)	0.953 (0.941–0.963)	0.736 (0.693–0.774)
	0.5	0.880 (0.861–0.898)	0.846 (0.812–0.880)	0.823 (0.786–0.856)	0.913 (0.894–0.933)	0.953 (0.941–0.963)	0.736 (0.691–0.775)
RF	0.35	0.856 (0.834–0.875)	0.743 (0.707–0.779)	0.926 (0.898–0.948)	0.815 (0.786–0.842)	0.958 (0.947–0.968)	0.741 (0.703–0.777)
	0.4	0.866 (0.844–0.885)	0.768 (0.730–0.804)	0.906 (0.876–0.932)	0.842 (0.814–0.866)	0.958 (0.947–0.968)	0.748 (0.708–0.786)
	0.45	0.884 (0.865–0.901)	0.810 (0.775–0.845)	0.890 (0.859–0.917)	0.880 (0.856–0.902)	0.958 (0.947–0.968)	0.770 (0.731–0.807)
	0.5	0.885 (0.866–0.902)	0.833 (0.799–0.867)	0.859 (0.825–0.889)	0.901 (0.879–0.921)	0.958 (0.947–0.968)	0.759 (0.718–0.796)

ANN, Hybrid, and RF models, highlighting the variations in classification trade-offs, with the confidence intervals (CIs) providing insight into the consistency of these estimates.

For the ANN model, as the threshold increases from 0.35 to 0.5, a slight increase in accuracy was observed (0.855–0.861) with the overlapping CIs (0.834–0.880), indicating no statistically meaningful differences in accuracy between thresholds. The sensitivity decreased (the model became less sensitive) from 0.836 (95 % CI: 0.801–0.867) at a threshold of 0.35 to 0.789 (95 % CI: 0.750–0.822) at 0.5, while the precision and specificity increased across the threshold. The model showed a constant AUC across the thresholds, and the best-balanced point between sensitivity and specificity was observed at the threshold

of 0.4, where the Youden index was slightly higher (0.704; 95 % CI: 0.660–0.746).

For the Hybrid model, as the threshold increased, the sensitivity (recall) decreased (0.874–0.823), while specificity and precision increased. The Youden index values were almost identical at the threshold of 0.35 and 0.4, which provided a balanced performance. At 0.35, the Hybrid model achieved an accuracy of 0.871 (95 % CI: 0.852–0.890) and improved at 0.4 with 0.873 (95 % CI: 0.853–0.892). Sensitivity was 0.874 (95 % CI: 0.841–0.902) and decreased to 0.4 with 0.861 (95 % CI: 0.827–0.890). Precision (0.794–0.846) and Specificity (0.870–0.913) increased as the threshold increased from 0.35 to 0.5. The overlapping confidence intervals for the Youden index indicate that the

slight differences between thresholds 0.35 and 0.4 are not statistically significant, as the model does not lose its ability to differentiate between the anaemic and not-anaemic cases, and provides a trade-off between sensitivity and specificity.

Additionally, the RF model showed a decline in sensitivity (0.926–0.859) across all thresholds from 0.35 to 0.5, making it less effective at predicting positive cases and minimizing false negatives. However, its performance in specificity (0.815–0.901) and precision (0.743–0.833) increased as the thresholds also increased. The RF model also showed consistency in the AUC with the high value (0.958) across all thresholds, representing the strong general discriminative power. The optimal threshold was observed at 0.45.

6.1.2. Literature benchmarking of the proposed model

Table 3 shows the benchmarked performance of the Hybrid machine learning model against the models used in previous studies. The performance of the evaluation metrics was presented at the standard threshold of 0.5, as reported in the previous works. The results are presented as the point estimates with 95 % CI for accuracy, precision, sensitivity, specificity, and AUC.

The Hybrid model achieved an accuracy of 0.880 (95 % CI: 0.861–0.898) and an AUC of 0.953 (95 % CI: 0.941–0.963). This was higher than the traditional models such as LASSO, SVM, Logistic Regression, and Naive Bayes, which showed a lower AUC ranging from 0.933 to 0.943. Additionally, the Hybrid model performed better than the classical models across most metrics, with higher CIs indicating a statistical difference. Also, the Hybrid model was stronger than the ANN model across all metrics. The RF and XGBoost models showed similar AUC (0.958, CI: 0.947–0.968, and 0.958, CI: 0.948–0.968), with overlapping CIs indicating no statistical differences. Compared with the RF model, the Hybrid model had a slightly lower sensitivity but better precision and specificity, leading to a balanced overall performance. Additionally, compared with the XGBoost model, the Hybrid model had a slightly lower accuracy (0.880 vs 0.892) and AUC (0.953 vs 0.958) but remained very close, especially in precision (0.846 vs 0.852) and specificity (0.913 vs 0.915). Additionally, the Hybrid model

6.1.3. Threshold optimization analysis

Fig. 3 shows the trade-off between sensitivity and specificity at different classification thresholds using Youden's J index for the RF, ANN, and Hybrid models. Each point on the graph represents a different threshold point, from 0.35 to 0.5. As the threshold increases, the sensitivity decreases and specificity increases, while decreasing the threshold, sensitivity increases and specificity decreases. Among all models, the RF model constantly reaches the highest sensitivity (recall), especially at the lower threshold of 0.35; however, it shows a decline in specificity. The hybrid model, at the threshold of 0.4, shows a more balanced performance, obtaining a nearly optimal trade-off between sensitivity (0.861) and specificity (0.880).

6.1.4. Statistical significance tests

The pairwise tests were conducted between the Hybrid model and the base models; McNemar's test was applied to examine whether the differences in misclassification errors between models were significant,

and DeLong's test was used to compare AUCs, ensuring that the observed differences reflected the actual performance variation rather than random chance. At the Youden-optimized thresholds, the McNemar's test showed that the Hybrid model made significantly fewer misclassification errors than ANN ($p = 0.0022$), indicating a statistical difference between the Hybrid model and the ANN model, while the test showed no statistical difference between the Hybrid model and the RF model ($p = 0.4885$). Additionally, the DeLong's test confirmed no significant differences in AUC between the Hybrid model and ANN (0.9381), as well as the Hybrid model and the RF model (0.9581), as shown in Table 4.

6.1.5. Error analysis

Error analysis was performed to identify patterns of false positives and false negatives, showing where the models failed or went wrong. This is particularly important in anaemia screening since false negatives (missed anaemic cases) carry greater clinical risk than false positives. Hence, making error analysis was essential for assessing the consistency and reliability of the models beyond the overall metrics, as shown in Table 5

For the Hybrid model, at the optimal thresholds of 0.35, the sensitivity (0.874, 95 % CI: 0.841–0.902) and specificity (0.870, 95 % CI: 0.845–0.892), corresponding to 101 false positives (FP) and 56 false negatives (FN) and at 0.4, the Hybrid model achieved the sensitivity (0.861, 95 % CI: 0.827–0.890) and specificity (0.880, 95 % CI: 0.856–0.902), corresponding to 93 false positives (FP) and 62 false negatives (FN). In comparison, the ANN model at a threshold of 0.4 produced more false positives (92 FN) than false negatives (79 FN). For the RF model, the lowest number of missed cases (49 FN) was observed, but with slightly higher false positives (93 FP) at threshold 0.45. Importantly, the Hybrid model deliberately accepts more false positives to reduce false negatives, which is clinically valuable for anaemia prediction, where missing an anaemic case has a higher cost than unnecessary treatment.

6.2. Discussions

The hybrid machine learning model was built using a stacked ensemble approach that combined the strengths of two base models: Random Forest (RF) and Artificial Neural Network (ANN), with a meta-learner model, XGBoost, that aggregated the predictive probabilities from the base models. According to Rane et al. (2024), this approach benefits from the strengths of ensemble learning and neural networks to improve generalisation and predictive accuracy and overcome the limitations of a standalone model. It operates at two levels. First, base learners are trained on the original dataset, and second, a meta-learner is trained on the outputs of the base models (predicted probabilities). Additionally, the RF model, due to its ensemble decision trees, can handle noise well and resist the overfitting problem, and the ANN model is capable of dealing with nonlinear relationships and patterns between the input and the target features due to its deep learning manner (Geethanjali & Valarmathi, 2024). Also, the meta-learner, XGBoost model, was chosen for its speed and performance, especially in predictive modelling competitions, and it's effective in handling several types

Table 3
Literature Benchmarking of the proposed Model.

Model	T	Accuracy	Precision	Sensitivity	Specificity	AUC
Logistic Regression	0.5	0.860 (0.839–0.880)	0.780 (0.744–0.817)	0.859 (0.825–0.889)	0.860 (0.836–0.886)	0.943 (0.930–0.953)
LASSO	0.5	0.859 (0.839–0.879)	0.780 (0.743–0.817)	0.857 (0.823–0.886)	0.860 (0.836–0.886)	0.943 (0.930–0.954)
SVM	0.5	0.861 (0.842–0.881)	0.814 (0.778–0.850)	0.805 (0.769–0.841)	0.894 (0.872–0.917)	0.934 (0.919–0.946)
Naive Bayes	0.5	0.859 (0.837–0.880)	0.811 (0.776–0.847)	0.800 (0.760–0.836)	0.893 (0.869–0.916)	0.933 (0.919–0.945)
Random Forest	0.5	0.885 (0.866–0.902)	0.833 (0.799–0.867)	0.859 (0.825–0.889)	0.901 (0.879–0.921)	0.958 (0.947–0.968)
ANN	0.5	0.861 (0.839–0.880)	0.822 (0.787–0.858)	0.789 (0.750–0.822)	0.902 (0.879–0.924)	0.934 (0.920–0.946)
XGBoost	0.5	0.892 (0.874–0.909)	0.852 (0.820–0.884)	0.852 (0.820–0.881)	0.915 (0.895–0.934)	0.958 (0.948–0.968)
Hybrid (RF+ANN→XGB)	0.5	0.880 (0.861–0.898)	0.846 (0.812–0.880)	0.823 (0.786–0.856)	0.913 (0.894–0.933)	0.953 (0.941–0.963)

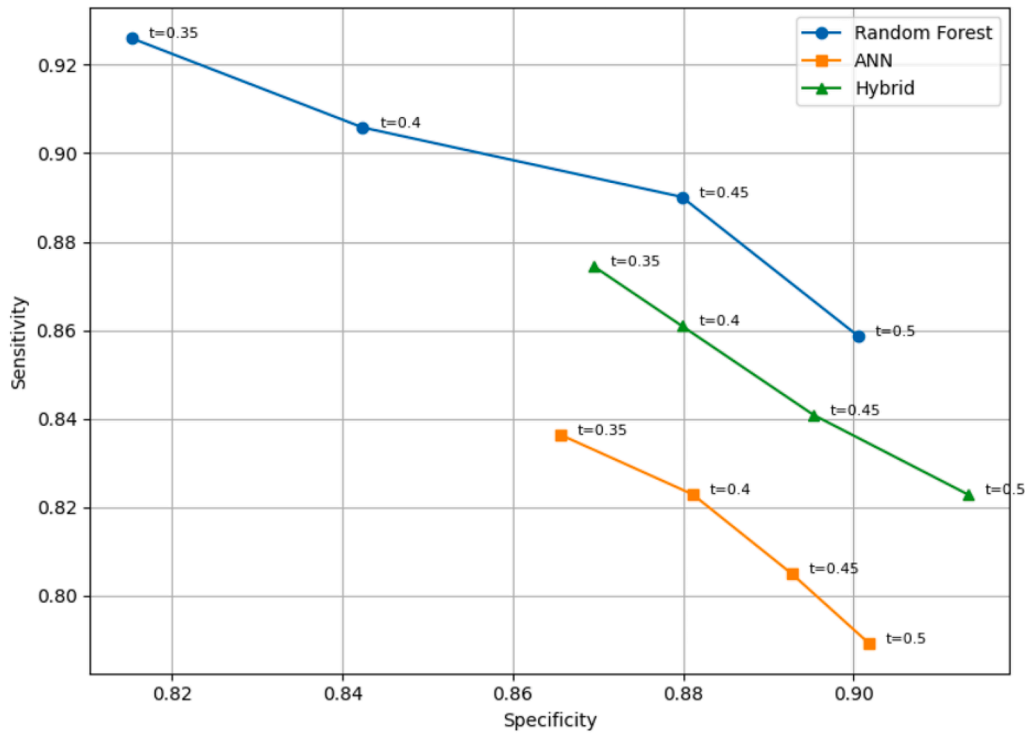


Fig. 3. Sensitivity-Specificity Trade-off.

Table 4
Statistical significance tests.

Comparison	McNemar b	McNemar c	McNemar p-value	DeLong AUC_H	DeLong AUC_Model	DeLong z	DeLong p-value
Hybrid vs RF	23	29	0.4885	6.579	0.022	0.953	0.9581
Hybrid vs ANN	41	17	0.0022	6.579	0	0.953	0.9342

Table 5
Error analysis.

Model	Threshold	Sensitivity (95 % CI)	Specificity (95 % CI)	TN	FP	FN	TP
ANN	0.35	0.836 (0.801–0.867)	0.866 (0.840–0.890)	670	104	73	373
	0.4	0.823 (0.786–0.855)	0.881 (0.858–0.904)	682	92	79	367
	0.45	0.805 (0.766–0.839)	0.893 (0.869–0.915)	691	83	87	359
	0.5	0.789 (0.750–0.822)	0.902 (0.879–0.924)	698	76	94	352
Hybrid	0.35	0.874 (0.841–0.902)	0.870 (0.845–0.892)	673	101	56	390
	0.4	0.861 (0.827–0.890)	0.880 (0.856–0.902)	681	93	62	384
	0.45	0.841 (0.805–0.872)	0.895 (0.874–0.917)	693	81	71	375
	0.5	0.823 (0.786–0.856)	0.913 (0.894–0.933)	707	67	79	367
RF	0.35	0.926 (0.898–0.948)	0.815 (0.786–0.842)	631	143	33	413
	0.4	0.906 (0.876–0.932)	0.842 (0.814–0.866)	652	122	42	404
	0.45	0.890 (0.859–0.917)	0.880 (0.856–0.902)	681	93	49	397
	0.5	0.859 (0.825–0.889)	0.901 (0.879–0.921)	697	77	63	383

of structured data (Ngo et al., 2022).

When evaluated the performance at the standard threshold of 0.5, the Hybrid model achieved an accuracy of 0.880 (95 % CI: 0.861–0.898), precision of 0.846 (95 % CI: 0.812–0.880), sensitivity of 0.823 (95 % CI: 0.786–0.856), specificity of 0.913 (95 % CI: 0.894–0.933), and AUC of 0.953 (95 % CI: 0.941–0.963). In comparison with the classical models, such as SVM, LASSO, Naive Bayes, and Logistic Regression, produced accuracies ranging from 0.859 to 0.860 and AUCs of 0.933 to 0.943 with weaker balance across sensitivity and specificity. These findings indicate that although the traditional models provide reasonable predictive performance, they tend to favor one evaluation metric over the other. Relative to the advanced single models, XGBoost and RF were the strongest, where XGBoost achieved

the highest accuracy (0.892) and AUC (0.958), while RF was close, achieving an accuracy of 0.885 and AUC of 0.958. The Hybrid model was nearly as good as XGBoost, just slightly lower in accuracy and AUC. But it had a better overall balance. Additionally, the Hybrid model outperformed the ANN model, representing the value of stacking. This design reflects the importance of combining different base learners to capture both the linear and nonlinear patterns, resulting in more stable predictions.

Benchmarking the Hybrid model against the broader literature supports these observations. The Hybrid machine learning model developed outperformed several existing models reported in the literature regarding balanced predictive performance, especially in an imbalanced dataset like health datasets. Although Uwizera et al. (2022) reported a

higher sensitivity of 98.71 % and an AUC of 98 % using an XGBoost model, but with a cost of specificity, the hybrid model in this study kept a more balanced trade-off between sensitivity and specificity (82.3 % and 91.3 %, respectively) and a strong AUC of 95.3 %. Related to the other studies, such as those by [Khan et al. \(2021\)](#) and [Tesfaye et al. \(2024\)](#), which state lower accuracies and specificity levels, the hybrid model showed greater strength and generalizability. While another study conducted by [Zahirzada et al. \(2023\)](#) attained a comparable accuracy of 86.4 %, the hybrid model showed a slightly improved overall balance across all metrics. These results suggest that combining multiple base learners with a stacking ensemble approach can enhance model stability and predictive power in class-imbalanced datasets such as health outcome prediction tasks ([Ngo et al., 2022](#); [Rane et al., 2024](#)).

While previous studies have limited evaluation to the default threshold of 0.5, this study introduces threshold optimization by evaluating thresholds of 0.35, 0.4, 0.45, and 0.5 and provides a more reliable understanding of trade-offs between sensitivity and specificity and improves accuracy. A study conducted by [Freeman & Moisen, 2008](#) reported that using a traditional default threshold of 0.5 as the cut-off does not necessarily preserve the observed prevalence or result in the highest prediction accuracy, especially for datasets with imbalanced problems (datasets with very high or very low observed prevalence). In addition, in many binary classification tasks, the goal is to optimize a single performance measure, for example, accuracy or sensitivity. But there are serious situations where it is necessary to maximize two or more performance measures simultaneously to ensure a balanced and robust classifier performance. For example, in medical diagnostics, sensitivity is needed to detect as many positive cases as possible, while specificity is necessary to avoid misdiagnosing healthy individuals ([Foody, 2023](#)). In this study, at the threshold of 0.4, the hybrid model showed a more favourable balance between sensitivity and specificity (86.1 % and 88.0 % respectively), which is necessary in medical screening like anaemia detection, where both false positives and false negatives have a significant cost in the life of the children aged 6–59 months.

From a clinical standpoint, false negatives represent the number of true anaemic cases classified as not anaemic, leading to delayed treatment, disease development, and avoidable health problems. This type of error carries the highest clinical cost. False positives represent the number of not-anaemic children being classified as anaemic cases, leading to unnecessary treatment and additional resource use, but are generally less harmful to the patient.

The trade-off between 0.35 and 0.4 of the Hybrid model carries practical implications. At 0.35, the model prioritizes detecting as many anaemic cases as possible (fewer FN), but with the cost of treating more healthy individuals (higher FP). At 0.4, the model becomes more suitable for reducing the number of false alarms related to healthcare resources and patient anxiety. However, there are a few true anaemic cases missing.

When compared with ANN, at the Youden-optimized threshold, the Hybrid model reduced the number of missed anaemic children (62 vs 79 FN), which is critical for timely intervention. While the RF model and the Hybrid model both performed better, the differences between them were not statistically significant. This suggests that the RF model appeared to be the best in terms of minimizing missed cases, the Hybrid model offered a more balanced trade-off between sensitivity and specificity, which is clinically safer, reducing missed diagnoses without overburdening the system with false alarms.

Overall, the sensitivity (recall) is critical, as missed anaemic cases (false negatives) have a greater risk than false positives. The Hybrid model achieved a sensitivity of above 86 % at a Youden-optimized threshold, with a balanced specificity around 88 %, providing a necessary trade-off between detecting true cases and minimizing false cases. The higher specificity of the Hybrid model reduces unnecessary treatment, while maintaining a strong sensitivity, ensuring fewer missed diagnoses, indicating the robustness of the Hybrid model to be considered as a practical decision support tool in screening and early detection

of anaemia in children, making the Hybrid approach coupled with the threshold optimization technique using Youden's index more suitable for real-world anaemia screening programs, especially in situations where both under-diagnosis and over-diagnosis carry a significant burden.

7. Conclusions and recommendations

7.1. Conclusion

To improve the generalizability and overcome the limitations of standalone models especially in dealing with imbalanced cases like health datasets, the study concludes to use hybrid machine learning model using a stacked ensemble approach combining the strengths of two base models, Random Forest (RF) and Artificial Neural Network (ANN), with a meta-learner, XGBoost model that aggregated the predictive probabilities from the base models. The hybrid machine learning model and the base models were evaluated for accuracy, precision, sensitivity, specificity, and Area Under the Curve (AUC) with 95 % confidence intervals (CIs) through different decision thresholds using Youden's J index (0.35, 0.4, 0.45, and 0.5). Additionally, statistical differences were assessed using McNemar's test to test whether one model outperforms the other, and DeLong's test was used to compare AUCs between ROC curves. Through threshold optimization, the RF model maintained its high sensitivity, but with lower precision and specificity while the hybrid model showed a consistently balanced trade-off between sensitivity and specificity across all thresholds, and the most favourable threshold was 0.4, which provided a more balanced sensitivity and specificity that minimized both false positives and false negatives in anaemia prediction for children aged 6–59 months. Although the RF model minimized the false negatives slightly more, the differences with the Hybrid model were not statistically significant. Clinically, the Hybrid model proved safer by ensuring fewer missed cases without overburdening the system with false cases. The findings highlight the robustness and practical relevance of the Hybrid model as a guiding tool for anaemia screening programs. Hence, the study presents a novel stacking hybrid model that integrates Random Forest (RF) and Artificial Neural Network (ANN) using XGBoost as a meta-learner, joined with threshold optimization.

7.2. Contributions

This study has made the following theoretical, methodological, and practical contributions.

This study enhances the theoretical understanding of childhood anaemia by identifying its multiple causes. It confirms that factors such as child-level, maternal characteristics, and household-related factors collectively affect the risk of anaemia, thereby recommending multi-dimensional health frameworks. Additionally, it also places global anaemia classification standards in the context of Tanzania. This contributes to the local implementation of global health guidelines, offering insights that bridge international health theory with national implementation.

Methodologically, this study presents a novel stacking hybrid model that integrates Random Forest (RF) and Artificial Neural Network (ANN) using XGBoost as a meta-learner, moving beyond the traditional single-model approaches. This model showed strong performance in addressing the class imbalance, a challenge in health data analysis. Additionally, integrating the threshold optimization techniques in classification tasks to balance sensitivity and specificity, thereby aligning the methodological design with clinical decision-making needs. The study also advances the methodological rigor by re-implementing and benchmarking the state-of-the-art classifiers on the same dataset, and by validating the performance improvements through confidence intervals and paired statistical significance tests.

In practical terms, this study provides a data-driven framework

designed to predict anaemia in children aged 6–59 months. The balanced performance of the hybrid model positions it well for incorporation into mobile health applications or public health dashboards. The study also illustrates the importance of national survey data for machine learning applications, promoting evidence-based decision-making and enhancing analytical capacity in low-resource environments. Additionally, this model provides a foundation for predictive modelling across several health domains, such as neonatal mortality, malnutrition, and infectious disease outbreaks.

7.3. Limitation

The study highly relied on secondary data that contained the social, demographic, and health-related factors. Although the dataset is nationally representative, it may lack certain socio-environmental features such as micronutrient intake and environmental exposure, which could further improve the model's predictive power and generalization. Additionally, the use of features such as breastfeeding duration, child birth size, and household water supply, which are not routinely recorded in the Health Management Information System (HMIS), limits the direct application of the model in real-world clinical settings.

7.4. Recommendations and future work

This study presents the following recommendations and outlines the future work.

- i. Health planners and policymakers also improve the campaigns on hygiene practices, safe water, and food preparation, and increase the deworming and immunization coverage to reduce infectious diseases.
- ii. Health planners and policymakers also should adopt data-driven approaches in designing and evaluating anaemia control programs, investing in strengthening health information systems to collect high-quality data for predictive modelling routinely.
- iii. The study also recommends deploying the hybrid model (RF+ANN with XGBoost as a meta-learner) as a decision-support tool in community health programs and clinics to identify children at risk of anaemia at an early stage.
- iv. Based on the relevant social, demographic, and health-related factors associated with anaemia in children aged 6–59 months, the study recommends that health planners and policymakers design and implement the appropriate interventions to eliminate anaemia.
- v. The study also recommends further related or longitudinal studies incorporating additional socio-environmental variables such as household dietary diversity and environmental exposure, to improve the model's generalizability and reliability.

Data statement

This study used secondary data from the Tanzania Demographic and Health Survey conducted in 2022. Data is publicly available. It can be accessed upon request from the National Bureau of Statistics (NBS), Tanzania.

Ethics statement

This study used only anonymized secondary data from the Tanzania Demographic and Health Survey (TDHS) 2022. No additional ethical approval was required. The TDHS 2022 protocols were reviewed and approved by the Tanzania National Institute for Medical Research (NIMR) and the Institutional Review Board (IRB) of ICF International.

Reproducibility

The code will be available upon request from the Corresponding

Author.

Author contributions

Rabia Omar Said and Mahadia Tunga participated in the conceptualization of the study. Rabia Omar Said performed data curation. Rabia Omar Said and Mahadia Tunga performed formal analysis. Rabia Omar Said and Mahadia Tunga designed the methodology. Mahadia Tunga supervised the study. Rabia Omar Said developed the model, and Mahadia Tunga performed validation. Rabia Omar Said wrote the original draft. Mahadia Tunga performed the review and editing of the paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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